MACHINE LEARNING OPERATIONS

MLOps

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Objectives

- Understand the difference between MLOps and traditional DevOps
- Design and evaluate the pipeline of a ML model and its operational data
- Architect the scaling of ML with platform

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Topics

- **Introduction to MLOps**
- Pipeline design
- Monitoring and logging
- MLOps with Cloud platforms

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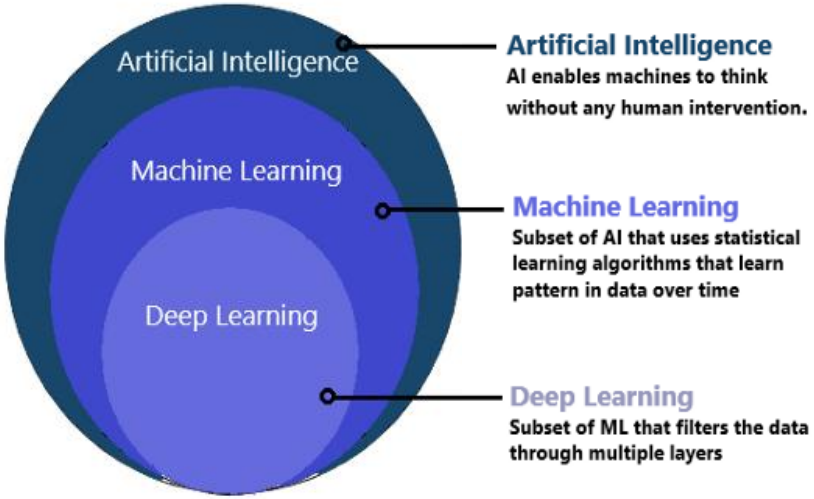
What is Machine Learning ?

- involves the use of ***algorithms*** and ***statistical*** models to enable computers to perform tasks without ***explicitly programmed***

- a subset of artificial intelligence (AI)



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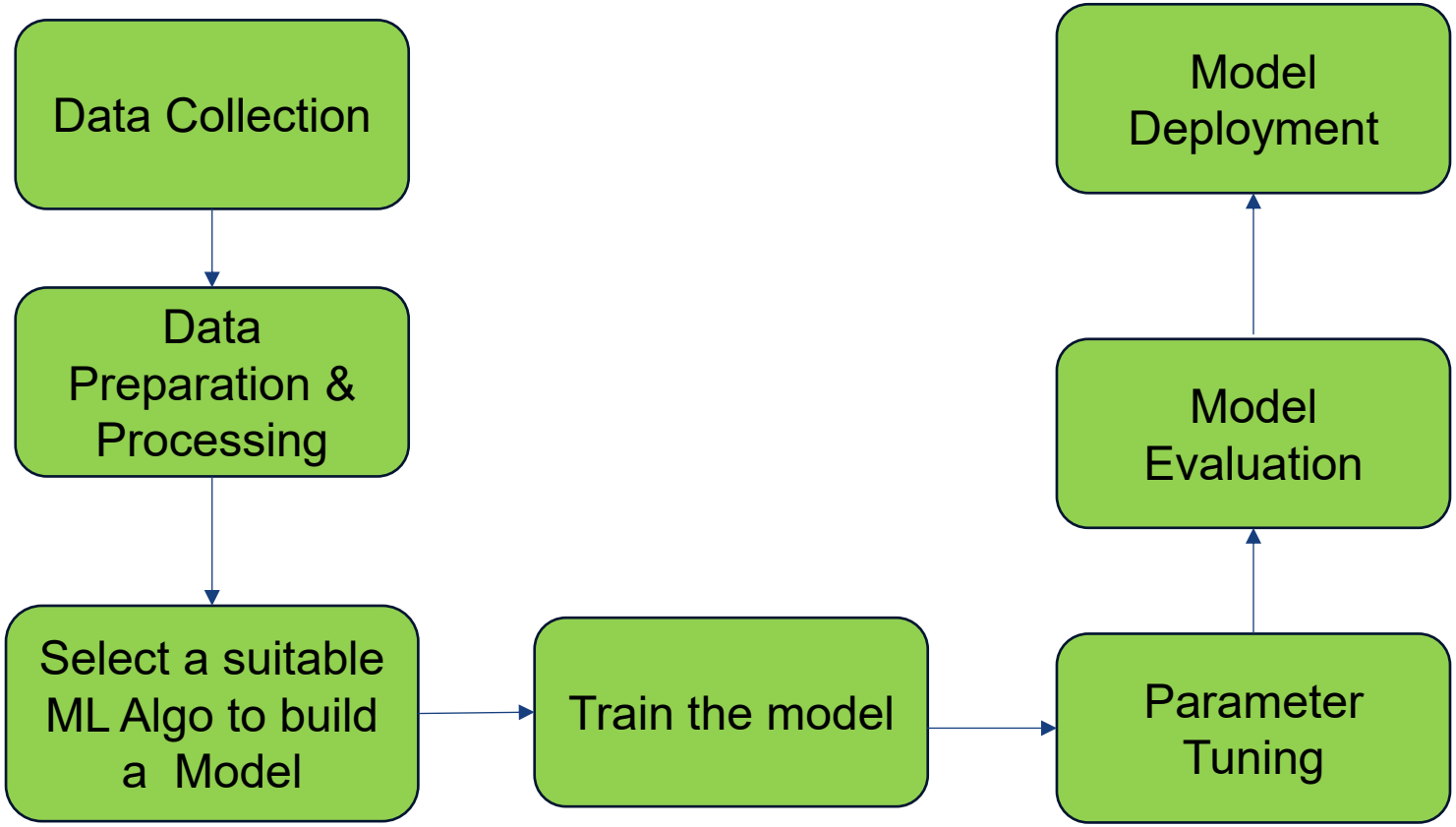
Life cycle of Machine Learning Model Development



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What is MLOps ?

Discipline of applying DevOps principles to:

- Lifecyle of ML models from development to production
- Streamline and automate the deployment, monitoring and management of ML models

Purpose:

- Continuous delivery of ML solutions
- Reliable and scalable

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Why MLOps ?

Scaling

- MLOps is essential for **massively deploying machine learning models**
- **Keep track of versioning** during the design phase, understand whether retrained models are better, and ensure best performing models performing are promoted to production
- Ensures **model performance is not degrading in production** at defined periods

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Why MLOps ?

Enhancing Transparency and Accountability

- **Accountability**: Enables centralized control, management, and auditing of enterprise AI efforts, providing an overall view of data usage and model application, and ensuring traceability if something goes wrong

- **Transparency**: Allows the information of what ML models are deployed and their effects on the business.

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Why MLOps ?

Collaboration Across Teams

- Foster strong collaboration between data scientists, ML engineers, and operations teams.

- Without collaboration, silos can cause:
  - ❖ Misalignment between model development and production constraints.
  - ❖ Slow deployment due to incompatible environments.
  - ❖ Poor monitoring or retraining triggers.

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Why MLOps ?

Continuous Improvement and Feedback loop

- **Keep track of versioning** during the design phase, understand whether retrained models are better, and ensure best performing models performing are promoted to production
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Introduction to MLOps

Key components of MLOps

- Model Development
- Model Deployment
- Monitoring
- Iteration
- Governance

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Model Development

- Data Engineering
- Feature Engineering (can involve feature stores*)
- Experiment Tracking (e.g. MLflow)
- Version Control Code and data versioning (e.g. Git) for reproducibility.
- Model Training Pipelines (e.g. Kubeflow, SageMaker Pipelines)

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Model Deployment

- Model Registry (e.g MLFlow)
- CI/CD for ML
- Serving Infrastructure (e.g. FastAPI, FlaskAPI)
- Containerization and Orchestration (e.g Docker and Kubernetes)

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Model Monitoring and Management

- Model Performance monitoring
- Data & Concept Drift Detection
- Logging and alert
- Model retraining pipelines

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Governance & Security

- Model Explainability
- Auditability
- Access Controls & Security
- Ethical & Bias Evaluation

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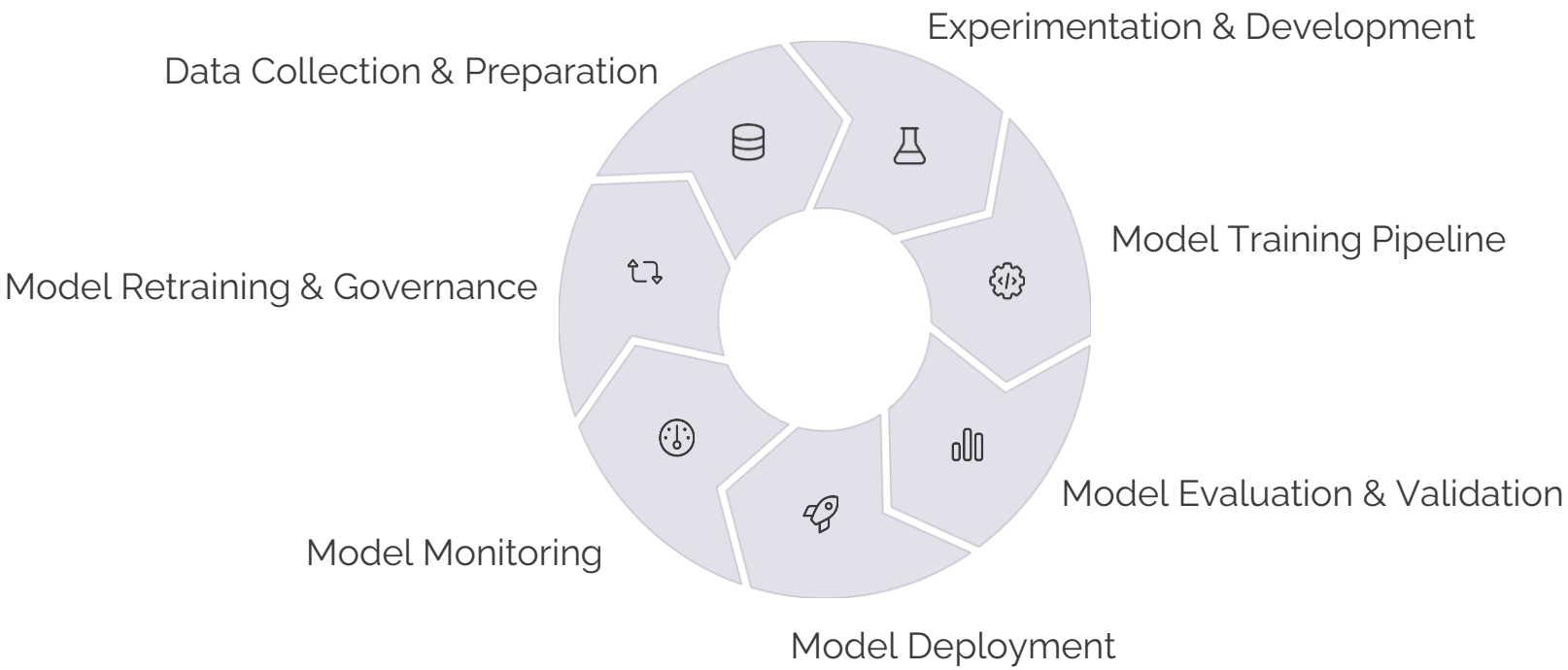
MLOps Lifecycle



The machine learning lifecycle is a continuous feedback loop where insights from monitoring drive improvements in future iterations.

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Data Collection and Preparation

Tasks:

- Data ingestion
- Cleaning, validation
- Labeling
- Feature engineering

Purpose:

- “Garbage in, garbage out” — ensure high-quality data is the foundation.

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Experimentation and Development

Tasks:

- Model selection
- Hyperparameter tuning
- Cross validation

Purpose:

- *Iterative model development and experimentation.*

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Model Training

Tasks:

- Batch or online training
- Pipeline automation
- Distributed training

Purpose:

- Scalable, reproducible training pipelines.

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Model Validation & Evaluation

Tasks:

- Perform error analysis (eg mean square error)
- Evaluate error score

Purpose:

- Ensure the model meets performance, robustness, and ethical standards.

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Model Deployment

Tasks:

- Build Rest API using framework like Fast API
- Dockerization

Purpose:

- Push the model into production with scalability and reliability

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Model Monitoring

Tasks:

- Monitor inputs & predictions
- Alert on degradation or anomalies
- Monitor system resource usage

Purpose:

- Track for degrading model performance

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Model Maintenance and Retraining

Tasks:

- Retraining with new data (scheduled or triggered)
- Model versioning and rollback
- Continuous learning or feedback loops

Purpose:

- Track for degrading model performance

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DevOps vs MLOps

Aspect	DevOps	MLOps
Focus	Software Application	ML Models, Intelligent Agent
Main Artifacts	Code + binaries	Code + models
Testing	Unit Testing, Integration Testing etc	Model validation +
Retraining Needs	No	Yes

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Key challenges

Challenge	Why It's Hard in ML
Data versioning	Code changes are easy; data changes are not
Experiment tracking	100s of runs per model, hard to reproduce
Deployment	Requires validation, explainability
Monitoring & drift detection	ML models degrade silently over time
Retraining automation	Triggering, testing, scaling retrains
Governance & fairness	Legal, ethical, and social implications
Team alignment	Different skills, different tools

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Model Reproducibility

- Model can be reproduced with the exact same results from scratch
- To ensure faithful reproduction in production
- Requires version control
- Tools like MLFlow greatly simplify the reproducibility implementation

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Model Scalability

Handling high-scale data in production

- Transition a few models to tens, hundreds, or even thousands
- Leveraging framework like Kubernetes simplifies the scaling of trained models by its auto-scaling feature
- Requires version control

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Model Scalability

Training large numbers of models

- Automated pipelines for model training
- Dynamic training parameters
- Experiment tracking
- Monitoring and feedback loop

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Ensuring the same data is available for reproducibility is crucial, though the storage capacity required to version data can be prohibitive given update rates and qua

Version control of ML Models

- Like software engineering **code versioning** matters for ML too
- **Data Versioning** to ensure the same data is available for reproducibility is crucial

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Model registry of ML Models

- **Centralized repository** specifically designed to manage the lifecycle of machine learning models
- Model registration will have unique name, version, and stage
- Allows **tracking of version history** of models, enabling the understanding of how models evolve
- Model stages (Development, Staging, Production)
- CI/CD Workflow Integration

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Summary.

MLOps applies DevOps principles to life cyle of ML models from development to production

Key components Model Development

- Model Deployment
- Monitoring
- Iteration
- Governance

7 MLOps lifecycle

MLOps Vs DevOps

Model Reproducibility and Scalability

Model versioning and registry

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